TAE KOAN YOO

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SELECTED PUBLICATIONS & PATENTS [Full Research Portfolio]

[1] [Preprint] **T. K. Yoo** et al., "PhysTrace: Interpretable Temporal Evaluation of Physical Consistency in Video Generative Models," arXiv preprint, 2026.
[2] [Pub.] **T. K. Yoo** et al., "PADS-TAL: Padding Annealed Diffusion Sampling in Text-Aware Latent Space for Robust and Diverse Text-to-Music Generation," *ICML,* 2026. (Presenter)
[3] [Pat.] **T. K. Yoo** et al., "Method and System for Generating Audio Source Incorporating User-Specified Melody Using a Diffusion Model," Korean Patent Application No. 10-2025-0134391, US Patent Application No. 19/810,102, 2025.
[4] [Pat.] J. H. Kim, H. C. Choi, J. H. Lim, **T. K. Yoo**, "Apparatus and Method for Object Recognition," US Patent Publication: US20240104935A1, 2024.
[5] [Pub.] **T. K. Yoo** et al., "VLSI Implementation of Area-Efficient Parallelized Neural Network Accelerator Using Hashing Trick," *ISOCC*, 2019. (Presenter)
[6] [Pub.] **T. K. Yoo** et al., "Analysis on Accelerating Object Detection in Edge Device with Half-Precision Floating Point," *KCC,* 2018. *Best Paper Award*. (Presenter)

RESEARCH & INDUSTRY EXPERIENCE [Related Research & Technical Writings]

INDEPENDENT RESEARCH (in collaboration with Prof. Kyeongbo Kong) **Apr. 2025 — Present**

- Introduced PhysTrace, a training-free framework for tracing physical consistency in video generative models via history-conditioned video–language alignment.
- Pursued independent research on alignment-preserving diversity control in diffusion models, developing PADS-TAL from problem formulation through publication at ICML 2026.

NHN CORPORATION, Seongnam, Republic of Korea **Sep. 2023 — Present**
Multi-modal AI Team, AI Tech Lab | *Senior Research Engineer*

- Led multimodal generative AI projects, developing NHN's first diffusion-based text-to-music service from proprietary data and adapting personalized text-to-image models for webtoons.
- Developed and optimized real-time automatic speech recognition and streaming pipelines.

HYUNDAI MOTOR COMPANY, Seoul, Republic of Korea **Jan. 2020 — Sep. 2023**
Autonomous Driving SW Team | *Research Engineer*

- Established Hyundai's first camera–LiDAR fusion model for 3D object detection using internal data, achieving the best internal performance and preparing it for production deployment.
- Designed a cross-attention-based camera–LiDAR fusion architecture to improve robustness to motion-induced misalignment, especially for long-range and small-object detection.
- Built and accelerated a production-stage monocular road-boundary recognition pipeline.

EDUCATION

SUNGKYUNKWAN UNIVERSITY, Seoul, Republic of Korea **Mar. 2011— Feb. 2020**

M.S., Electrical and Computer Engineering (advised by Prof. Jong Tae Kim) GPA: 4.04/4.5 (3.91/4)[1]

- Conducted research on efficient deep learning for embedded systems, including the development of Double-Stage Weight Sharing as my thesis to reduce model size while preserving accuracy.
- Served as a teaching assistant for undergraduate courses, supporting labs and assignments.

B.S., Electronic and Electrical Engineering UP-GPA: 4.26/4.5 (3.93/4)[1]

- National scholarship for academic excellence, 4 semesters.
- Investigated OS–storage optimization, proposing fTrim/vTrim schemes for the Linux filesystem–SSD interface. *Best Undergraduate/Junior Paper Award*.

** Undergraduate Research Student (advised by Prof. Dong Kun Shin) Aug. 2017– Feb. 2018
** Exchange Student Program, The University of Texas at Austin, USA Jan. 2017– Aug. 2017

[1] 4.0-scale GPA was converted using Scholaro.